The Role Family — Roles, WAR, and Emergent Archetypes

ABRA · Version 2.6.0 · 2026-07-24 · Will Hooper

A single read-through of the newest work: why the old single-label archetypes were wrong, how the role model fixes the data, what WAR measures, and what the emergent (NMF) archetypes actually found — with every result stated against an honest baseline.

---

1. The problem this fixes

The earlier model gave each team exactly **one** archetype label (Sun *or* Tailwind *or* Trick Room). That is a *multi-class* framing of a *multi-label* object: a real Champions team is Sun **and** Tailwind **and** Fake Out at once. Forcing one label throws away most of what a team is, and it shatters the data into archetype-by-archetype cells of only **11–18 games each** — which is exactly why those matchup numbers were untrustworthy and the "rock-paper-scissors" cycle had error bars crossing 50%.

The literature is explicit about the right framing: multi-label classification (Tsoumakas & Katakis, 2007), team-as-a-mixture-of-latent-roles (topic models; Blei, Ng & Jordan, 2003), and the finding that latent roles beat raw identity for outcome prediction in team sports (arXiv 2304.08272).

2. The role model

We define **26 functional roles** — speed control (Tailwind, Trick Room, speed drops), weather, terrain, disruption (Fake Out, redirection, Taunt, Encore), status, debuff (Intimidate and stat drops), priority, prankster, setup, healing, screens, walls, pivot, trapping, Perish, ally-support, item-disruption, and physical / special attacker.

Two design rules keep it honest:

- **Roles are earned from data.** A species is credited with a role only once it has actually been observed performing it (at least twice) across the store. This is "all the roles a Pokémon could play," learned rather than guessed.
- **Presence is binary; strength is learned, never typed.** A multi-effect move carries several *factual* roles (Matcha Gotcha = attack + heal + status; Body Press = wall + attack; Knock Off = attack + item strip; Fake Out = tempo, **not** an attacker). An earlier draft assigned fractional weights (0.6, 0.4...) by hand — those were removed, because a made-up number is asserted, not measured. The graded strength now comes out of the NMF (Section 4).

A team's role vector is built from the **team-preview six**, which are public in every closed-sheet game, so the representation does not leak and is not censored by who won.

Result — the pooling fix. Every game now contributes to many role-pair cells, so the median matchup cell rises from **n ≈ 15 to** n = 20 **across 1,051 cells (measured 2026-07-25 on 1,061 quality-filtered games)** — the 7,971 figure published in v2.6.0 was retracted in 2.7.0 as an over-tagging artifact and has since gone 7,971 → 95 → ~50 → 20, each with a Wilson confidence interval. That is the structural repair of the grid. But predicting the winner from preview roles still **ties a coin** (held-out log-loss 0.694

vs 0.693) — so the role model *describes and attributes*; it does not predict. The per-role logistic coefficients are read as **win-credit per role**, and KO-credit per species is measured directly from the turn log.

3. WAR — Wins Above Replacement

To attribute wins to individual Pokémon while controlling for teammates and opponents, we borrow basketball's **Regularized Adjusted Plus-Minus (RAPM)**. One row per game, label 1 if player-1 won, features = the difference of the two teams' species-presence vectors at preview. A ridge-regularized logistic regression gives each species an adjusted win contribution β; ridge shrinks rare species toward zero so a three-game fluke cannot post a huge number. With replacement set at the 20th-percentile β and the logistic slope of ¼ at a coin flip:

> ***WAR = 0.25 × (β − β_replacement) × games appeared.***

Result — WITHDRAWN. The species model appeared to beat a coin at held-out log-loss 0.6875 vs 0.6931, a figure **withdrawn 2026-07-25** — that figure was measured on the UNFILTERED store; on quality-filtered games WAR scores 0.7048 against a coin's 0.6931 (accuracy 0.502). The apparent signal was four bot accounts playing one team in 1,446 games — and beats the rating baseline (0.6905). So *which specific species* you bring at preview carries a small, real signal that roles alone and raw sheets do not. Leaders: Basculegion, Kingambit, Sylveon; trailers negative (Maushold, Raichu). Effect sizes are small and the magnitudes are ridge-shrunk, so WAR is an exploratory ordering, not settled wins.

4. Emergent archetypes (NMF)

Rather than hand-declaring roles, we let them fall out of the data with **Non-negative Matrix Factorization** (Lee & Seung, *Nature* 1999): approximate the big table as X ≈ W·H with everything non-negative, so each team is a **blend** of latent roles and each role is a recipe over features. Because nothing is negative, a team reads as "60% Intimidate control + 30% Tailwind offense," never as one minus another. A move's loading on a role is **learned**, which is the principled source of the graded strength we refused to type by hand (Label Distribution Learning; Geng, 2016).

Two cuts:

- **Team × move usage** (weighted by real in-battle usage, which down-weights the closed-sheet censoring bias) recovers **offensive cores** but is dominated by attacking moves — reconstruction error 0.79.
- **Team × role** recovers **six clean archetypes** — reconstruction error **0.53**:

#	Archetype	Composition (top roles)	Share
A1	Intimidate control	Debuff/Intimidate 56% · Fake Out 19% · Pivot 5%	20%
A2	Physical offense	Physical attacker 83% · Pivot 5% · Item disruption 5%	19%
A3	Special offense + sustain	Special attacker 81% · Status 7% · Healing 5%	18%
A4	Bulky wall + support	Wall 49% · Screens 9% · Redirection 8% · Healing 5%	17%
A5	Tailwind offense	Tailwind 85% · Encore 9%	15%
A6	Priority offense	Priority 89% · Setup 4% · Fake Out 3%	11%

The support/redirection core (A4) — the interesting, non-obvious structure — separates out cleanly here, which the move-level cut could not surface. The only human choices are the **rank** (six) and the **names**; everything else is from ~13,000 team-sides across the store.

5. Honest limits

- Preview composition barely separates from a coin. Role-level winner prediction ties it; WAR only edges it. The game is decided in play, not at preview — consistent with ABRA's central finding.
- Role tags are a **censored lower bound** on capability: closed sheets reveal only the moves that were actually clicked.
- NMF factors are soft, and at the move level attacker roles dominate. Reconstruction error is **not** comparable across different weightings; the correct model-selection criterion is **topic coherence** (Mimno et al., 2011), which is the noted next refinement rather than something already done.

Sources

Tsoumakas & Katakis 2007 (multi-label) · Blei, Ng & Jordan 2003 (LDA) · Lee & Seung 1999 (NMF) · Geng 2016 (Label Distribution Learning) · Mimno et al. 2011 (topic coherence) · Rosenbaum-style RAPM (basketball adjusted plus-minus) · latent roles in team sports (arXiv 2304.08272).